

# Task 2: House Price Prediction using Multiple Regression Models

## Internship Project Report:

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## Introduction:

This project predicts house prices using the California Housing Dataset by comparing Linear Regression, Ridge Regression, and Decision Tree Regression models. The models are evaluated using RMSE and  $R^2$  Score to identify the best-performing algorithm.

## Objective:

- Analyze the California Housing Dataset.
- Train and compare different regression models.
- Evaluate model performance using RMSE and  $R^2$  Score.

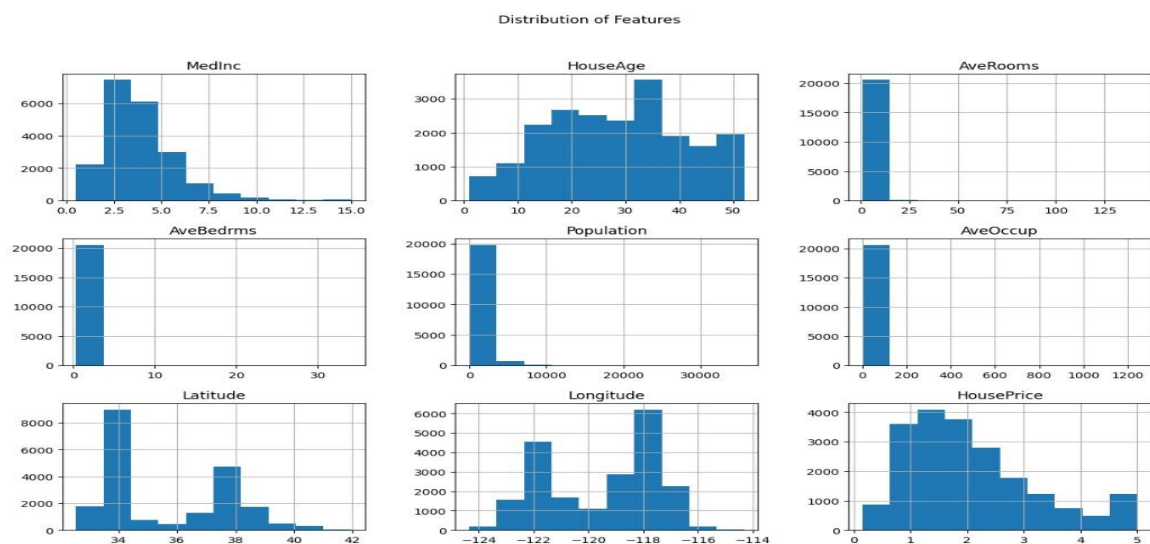
## Dataset Preview:

|   | MedInc | HouseAge | AveRooms | AveBedrms | Population | AveOccup | Latitude | Longitude | HousePrice |
|---|--------|----------|----------|-----------|------------|----------|----------|-----------|------------|
| 0 | 8.3252 | 41.0     | 6.984127 | 1.023810  | 322.0      | 2.555556 | 37.88    | -122.23   | 4.526      |
| 1 | 8.3014 | 21.0     | 6.238137 | 0.971880  | 2401.0     | 2.109842 | 37.86    | -122.22   | 3.585      |
| 2 | 7.2574 | 52.0     | 8.288136 | 1.073446  | 496.0      | 2.802260 | 37.85    | -122.24   | 3.521      |
| 3 | 5.6431 | 52.0     | 5.817352 | 1.073059  | 558.0      | 2.547945 | 37.85    | -122.25   | 3.413      |
| 4 | 3.8462 | 52.0     | 6.281853 | 1.081081  | 565.0      | 2.181467 | 37.85    | -122.25   | 3.422      |

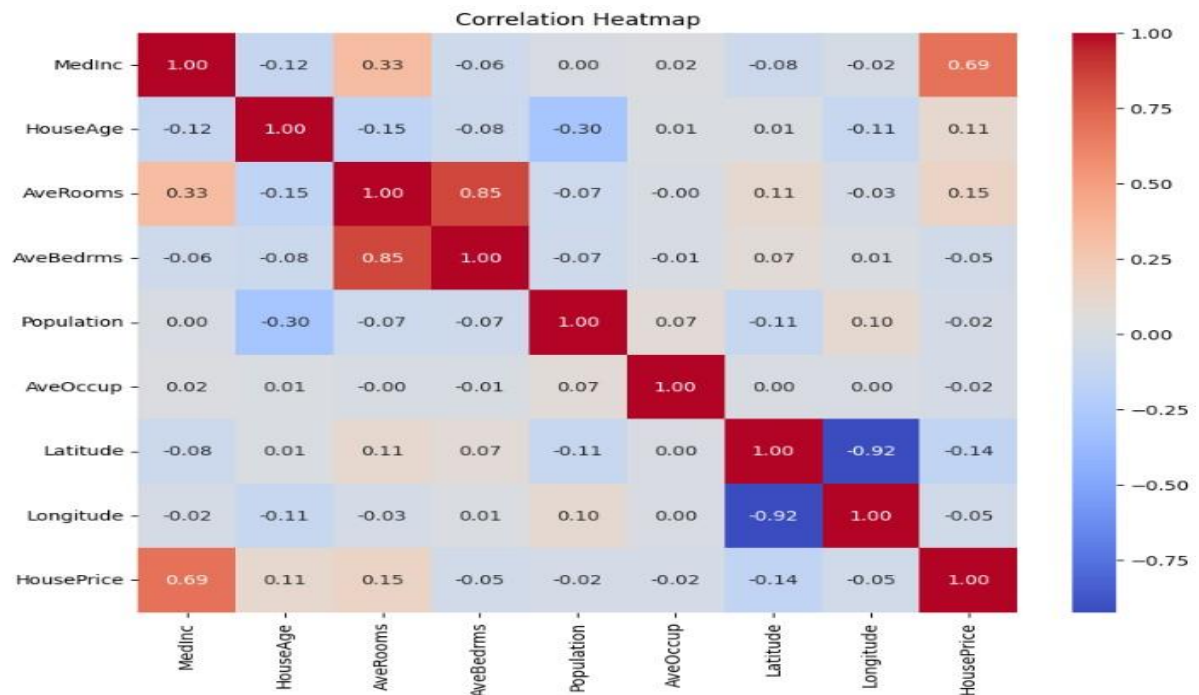
## Statistics:

|       | MedInc       | HouseAge     | AveRooms     | AveBedrms    | Population   | AveOccup     | Latitude     | Longitude    | HousePrice   |
|-------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| count | 20640.000000 | 20640.000000 | 20640.000000 | 20640.000000 | 20640.000000 | 20640.000000 | 20640.000000 | 20640.000000 | 20640.000000 |
| mean  | 3.870671     | 28.639486    | 5.429000     | 1.096675     | 1425.476744  | 3.070655     | 35.631861    | -119.569704  | 2.068558     |
| std   | 1.899822     | 12.585558    | 2.474173     | 0.473911     | 1132.462122  | 10.386050    | 2.135952     | 2.003532     | 1.153956     |
| min   | 0.499900     | 1.000000     | 0.846154     | 0.333333     | 3.000000     | 0.692308     | 32.540000    | -124.350000  | 0.149990     |
| 25%   | 2.563400     | 18.000000    | 4.440716     | 1.006079     | 787.000000   | 2.429741     | 33.930000    | -121.800000  | 1.196000     |
| 50%   | 3.534800     | 29.000000    | 5.229129     | 1.048780     | 1166.000000  | 2.818116     | 34.260000    | -118.490000  | 1.797000     |
| 75%   | 4.743250     | 37.000000    | 6.052381     | 1.099526     | 1725.000000  | 3.282261     | 37.710000    | -118.010000  | 2.647250     |
| max   | 15.000100    | 52.000000    | 141.909091   | 34.066667    | 35682.000000 | 1243.333333  | 41.950000    | -114.310000  | 5.000010     |

## Distributions:



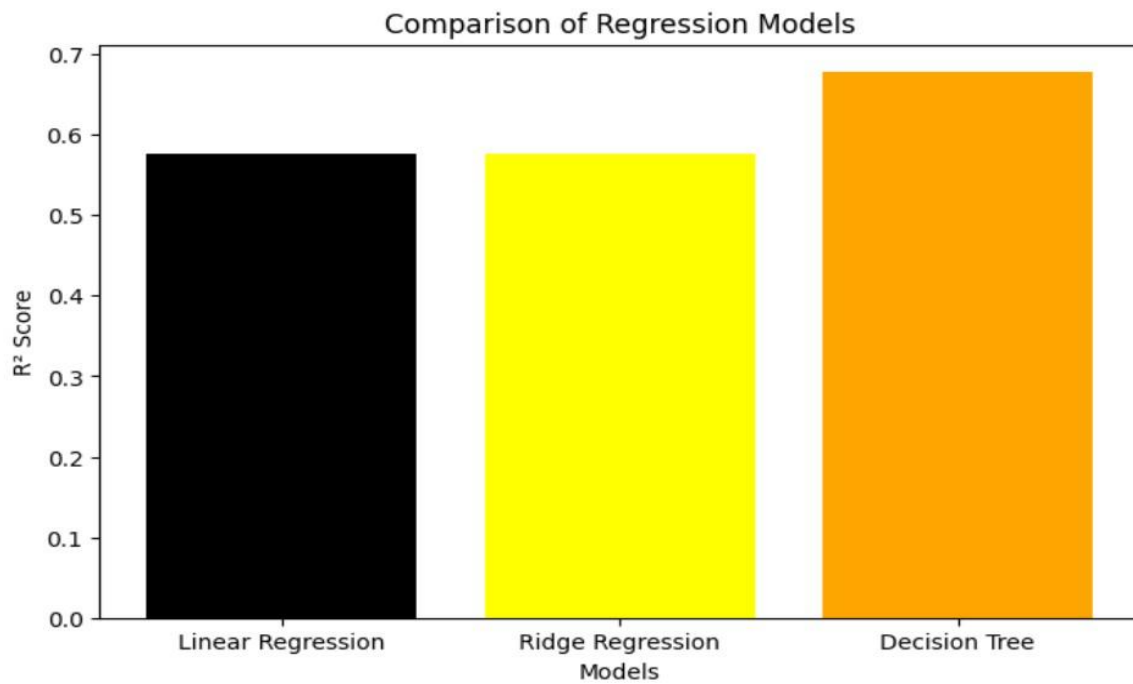
## Heatmap:



## Comparison Table:

|   | Model             | RMSE     | R2 Score |
|---|-------------------|----------|----------|
| 0 | Linear Regression | 0.745581 | 0.575788 |
| 1 | Ridge Regression  | 0.745557 | 0.575816 |
| 2 | Decision Tree     | 0.649650 | 0.677928 |

## Comparison Chart:



## **Conclusion:**

Decision Tree achieved the highest  $R^2$  score (~0.678), outperforming Linear and Ridge Regression (~0.576).